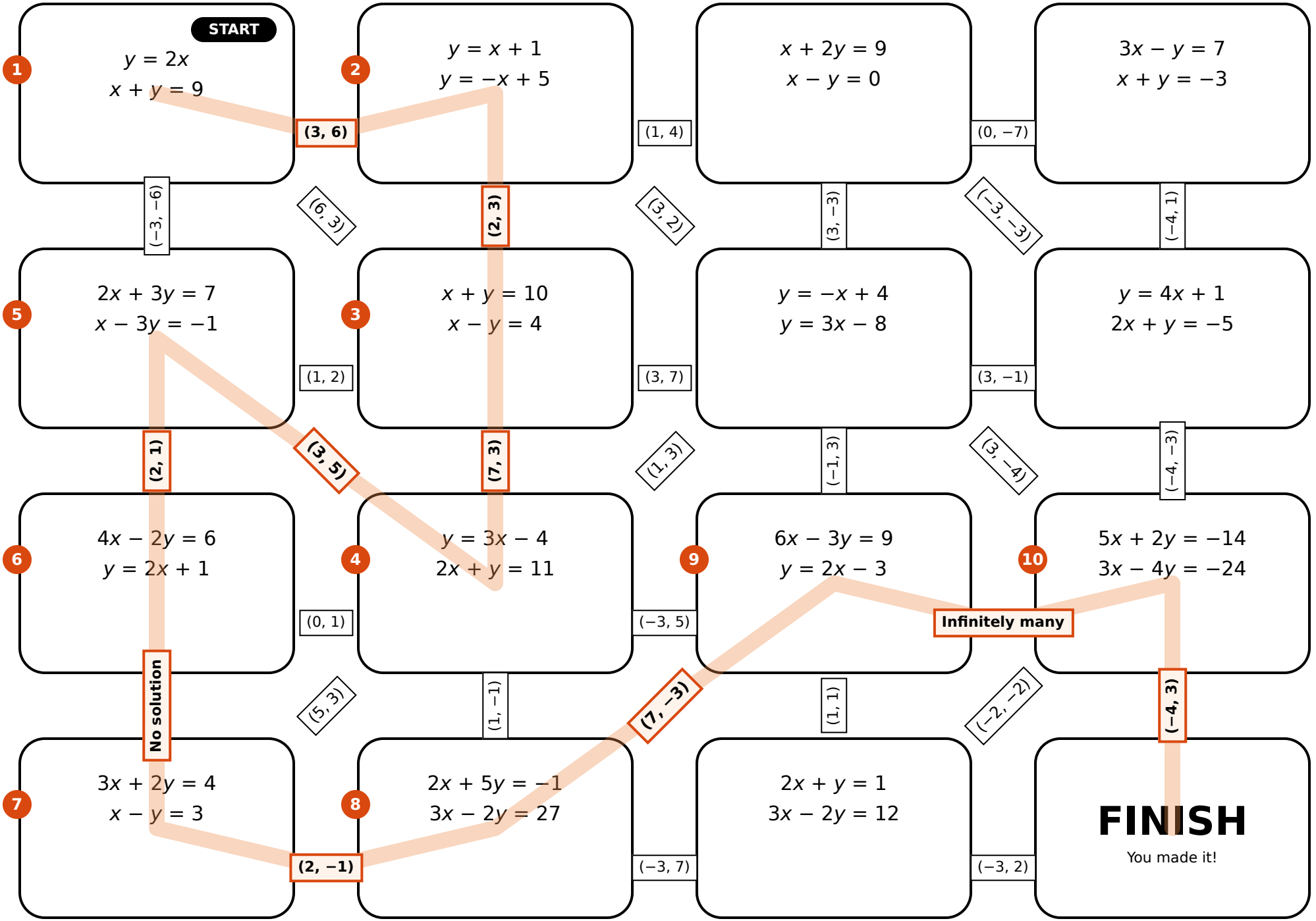
Systems of Equations Maze ANSWER KEY

Start in the top-left room. Solve the system and write the solution as a point (x, y) or as No solution / Infinitely many, then go through the door with your answer.



Systems of Equations Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **(3, 6)**	$x + 2x = 9 \rightarrow x = 3 \rightarrow y = 2(3) = 6$
2	Answer: **(2, 3)**	$x + 1 = -x + 5 \rightarrow 2x = 4, x = 2 \rightarrow y = 2 + 1 = 3$
3	Answer: **(7, 3)**	add: $2x = 14 \rightarrow x = 7 \rightarrow 7 + y = 10, y = 3$
4	Answer: **(3, 5)**	$2x + 3x - 4 = 11 \rightarrow 5x = 15, x = 3 \rightarrow y = 3(3) - 4 = 5$
5	Answer: **(2, 1)**	add: $3x = 6 \rightarrow x = 2 \rightarrow 2(2) + 3y = 7, y = 1$
6	Answer: **No solution**	$4x - 2(2x + 1) = 6 \rightarrow -2 = 6 \rightarrow$ false: parallel lines
7	Answer: **(2, −1)**	$x = y + 3 \rightarrow 3(y + 3) + 2y = 4 \rightarrow 5y = -5, y = -1 \rightarrow x = -1 + 3 = 2$
8	Answer: **(7, −3)**	$4x + 10y = -2$; $15x - 10y = 135 \rightarrow 19x = 133, x = 7 \rightarrow 14 + 5y = -1, y = -3$
9	Answer: **Infinitely many**	$6x - 3(2x - 3) = 9 \rightarrow 9 = 9 \rightarrow$ true: same line
10	Answer: **(−4, 3)**	$10x + 4y = -28$; $3x - 4y = -24 \rightarrow 13x = -52, x = -4 \rightarrow -20 + 2y = -14, y = 3$